KO LIH HAN

Malaysia | +60 17-424 6282 | lihhanko526@gmail.com | GitHub | LinkedIn

SOFTWARE ENGINEERING EXPERIENCE

Software Developer Aug 2022 – Aug 2026
Kahap Inc | Part-time

- Built and shipped a feedback platform covering submissions, discussions, admin review, and notifications.
- Kept the Laravel/Vue.js system running in production, handling queues, scheduled jobs, migrations, and Linux server operations.
- Added GPT-SoVITS, pronunciation checks, and text validation to backend speech workflows.

Full-Stack Developer Intern Jun 2021 – Jan 2022
Pac Com Solution

- Implemented features and fixes for a production Laravel/Vue.js POS system.

RESEARCH EXPERIENCE

Interpreting Implicit Intent through Workplace Hierarchies and Context

M.S. Thesis, National Tsing Hua University | NLP, Graph Learning, Role/Context KGs, Llama 3.1-8B

- Examined workplace utterances whose intent shifts with conversational context and speaker-listener roles, instead of interpreting each utterance on its own.
- Encoded LLM-extracted speaker/listener states with GCN/GAT, then used Role KG and Context KG retrieval to provide evidence for Llama 3.1-8B.
- On 400 target-only cases, DualKG reached semantic F1 0.682 versus 0.644 for the text-only baseline, outperforming Prior KG, Role KG, and Context KG.

PROJECTS

Small-Agent QLoRA

Python, Qwen3.5-9B, QLoRA / PEFT, Transformers, llama.cpp

- Built a tool-using Qwen agent with search, file inspection, Python execution, timeouts, duplicate blocking, and trace logs.
- QLoRA on corrective failure examples raised success from 34.5% to 62.0% on a 200-task evaluation.
- GAIA100 accuracy rose from 9% to 18%; trace review showed over-search was still the main failure mode.

Hybrid Retrieval Evaluation

Python, BM25, Qwen embeddings, RRF, FastAPI

- Combined BM25 and Qwen dense retrieval with RRF, while keeping rank provenance inspectable.
- Across 64 queries and 5,189 documents, Recall@10 rose from 0.734 to 0.763 and MRR@20 from 0.679 to 0.742.

Semantic Text-to-SQL

Python, FastAPI, LangGraph, SQLite, Qwen3.5-9B, pytest

- Built an NL-to-SQL pipeline with schema grounding, validation, bounded repair, and read-only execution.
- Kept FastAPI, CLI, Streamlit, and evaluation on one service layer to avoid behavior drift between interfaces.
- On 100 BIRD DEV queries, correct-query accuracy rose from 33% to 39%, fixing 6 cases without regressions.

EDUCATION

National Tsing Hua University Jun 2026
M.S., Information Systems & Applications | GPA 4.09/4.3

Chung Hua University Jun 2024
B.S., Computer Science & Information Engineering | GPA 3.9/4.0

SKILLS

AI / LLM:	Agents & tool use, RAG / hybrid retrieval, evaluation, PEFT / QLoRA, Transformers, LangGraph, llama.cpp
Backend:	Python, FastAPI, PHP, Laravel, REST APIs, SQL, MySQL, SQLite
Web & Tools:	JavaScript, Vue.js, Linux, Git, pytest
Languages:	Chinese (Native), English (Fluent), Malay (Intermediate)